# Group 19

*Credit Card Fraud Detection Project*

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- Progress Status Update

# Credit Card Fraud Detection

*Project Overview*

- Project focus: credit card fraud detection using machine learning
- Main workflow is complete up to model preparation
- Current status: in progress / mid-development

# Completed Work

*What is already done*

- Environment setup and dependency installation completed
- Dataset loaded successfully from data/creditcard.csv
- Data validation checks completed for required columns and class labels
- Exploratory analysis completed: class distribution, amount analysis, time patterns
- Feature engineering completed with time-based variables
- Train/validation/test split implemented chronologically

# Pending Work

*What remains to finish*

- Actual LSTM training loop still needs to run
- Model evaluation on validation and test data pending
- Metrics not yet produced: precision, recall, F1, ROC-AUC, PR-AUC
- No final performance summary or model comparison yet
- Result visualization and final conclusions still to be completed

# Current Status Summary

*Overall project review*

- The data pipeline, preprocessing, feature engineering, and model setup are complete
- The project is structurally ready for model training and evaluation
- The key remaining gap is the final training/evaluation phase
- The project is therefore in an intermediate but promising stage
- Next milestone: train final model and report results